

8. Relational database design

DATE: _____

PAGE: _____

2 →

$A \rightarrow BC$

(A determines B & C)

given value of A
we get unique value
of B & C

A	B	C
1	x	y
2	a	b
3	c	d

- This is called functional Dependency

ex.

stu-id, stu-name, stu-address, course-name, prereq, marks, grade are the attributes.

1. $\text{stu-id} \rightarrow (\text{stu-name}, \text{stu-address})$
2. $\text{course-name} \rightarrow \text{prereq}$
3. $\text{marks} \rightarrow \text{grade}$

2 →

Functional Dependency

A	B
1	3
1	5
2	6
3	7

$A \rightarrow B$ X

$B \rightarrow A$ V

A	B
1	3
1	3
2	6
3	7

$A \rightarrow B$ V

$B \rightarrow A$ V

schema
 $R(A, B)$

let R be a relational schema where $\alpha \subseteq R$,
 $\beta \subseteq R$ are attributes, then functional depend-
 $\alpha \rightarrow \beta$ holds if and only if, whenever
any two tuples t_1 & t_2 agree on α (same

values) they must agree on β .

purpose $\rightarrow$ to identify Redundancy and remove it.

$\rightarrow$ An attribute is not required for to be a key for functional dependency.

* Clouser set of Functional Dependency

$\rightarrow$ set of all F.D that are possible on Schema

$\rightarrow$ set F of F.D. (Given) $A \rightarrow B$
From this, $B \rightarrow C$
 $\Downarrow$
 F^+ (Additional F.D.) $A \rightarrow C$
 $\uparrow$
clouser set

$\rightarrow$ Armstrong's Axioms : (to find clouser set)

(1) $\alpha \Rightarrow$ set of attributes
 $\beta \Rightarrow$ subset of α ($\because \beta \subseteq \alpha$)
then $\alpha \rightarrow \beta$ holds

ex $AB \rightarrow A$

- This is called "trivial functional depⁿ".

- Known as "Reflex Intity".

Rule is

②

Augmentation : $A \rightarrow B \Rightarrow A \rightarrow B \vee C$

If $\alpha \rightarrow \beta$ holds and γ is also an attribute
 $\gamma\alpha \rightarrow \gamma\beta$ is also holds.

③

Transitivity :

If $\alpha \rightarrow \beta$ holds and $\beta \rightarrow \gamma$ holds then
 $\alpha \rightarrow \gamma$ will also holds.

Additional
rules :

④

Union :

If $\alpha \rightarrow \beta$ holds and $\alpha \rightarrow \gamma$ holds
then $\alpha \rightarrow \beta\gamma$ holds.

⑤

Decomposition : (Opposite of Union)

If $\alpha \rightarrow \beta\gamma$ holds then
 $\alpha \rightarrow \beta$ and $\alpha \rightarrow \gamma$ also holds.

⑥

Pseudo transitivity :

If $\alpha \rightarrow \beta$ holds and $\beta\gamma \rightarrow \delta$ holds then
 $\alpha\gamma \rightarrow \delta$ holds.

proof :

If $\alpha \rightarrow \beta$

$\Rightarrow \alpha\gamma \rightarrow \beta\gamma$

$\Rightarrow \alpha\gamma \rightarrow \delta$

Que: $A \rightarrow B$ $] \Rightarrow [A \rightarrow B]$ (9)

$A \rightarrow C$

$CC \rightarrow H$ $] \Rightarrow [CC \rightarrow H]$

$CC \rightarrow I$ $] \Rightarrow [A \rightarrow H]$

$B \rightarrow H$

if $A \rightarrow C$

$ACC \rightarrow CC$ and $CC \rightarrow I$

$\Rightarrow [ACC \rightarrow I]$

2 $\rightarrow$ Closure set of attributes

$A \rightarrow B$

$B \rightarrow D$

$C \rightarrow DE$

$CD \rightarrow AB$

closure of $A^+ = \{A, B, D\}$

$C^+ = \{C, DE, A, B\}$

Que

$A \rightarrow B$

$A \rightarrow C$

$A^+ \Rightarrow ACBCHT$

~~$C \rightarrow H$~~

~~$C \rightarrow I$~~

$B \rightarrow H$

Key attribute (because it gives all the attributes).

Algo

2 $\rightarrow$

α^+

result = α ; (Copy α in result)

repeat

for each F.D. $\beta \rightarrow \gamma$

begin

if $\beta \subseteq \text{result}$

result = result $\cup \gamma$

= $AC \cup B = ACB$

end

(For above ex. $\beta = A$ and $\gamma = B$)

until (result does not change)

$$\begin{aligned} \text{result} &= A \cup B = A \cup B \\ &= A \cup B \cup C = A \cup BC \\ &= A \cup BC \cup H = A \cup BCH \\ &= A \cup BCH \cup I = A \cup BCHI \end{aligned}$$

2 $\rightarrow$ $A \cup I^+ \Rightarrow$ Key, super key.

2 $\rightarrow$ is $A \cup I^+$ is candidate key $A^+ \Rightarrow BCH$
 $I^+ \Rightarrow I$.

is subsets of $A \cup I^+$ (BCH and I) gives us
super key $\Rightarrow$ no. $\Rightarrow$ therefore it is c.k.

$\Rightarrow$ it is one of the candidate keys.

2 $\rightarrow$ Use of ^{this} Algoⁿ: ($A \cup I^+$ closure algoⁿ)

- to test if α is a superkey.
($\alpha^+ \rightarrow$ gives everything $\Rightarrow$ s.k.)

- to check a F.D. $\alpha \rightarrow \beta$ holds.
(if α^+ gives us β , we can say $\alpha \rightarrow \beta$ holds)

- to compute f^+ (closure of F.D.)
(set of all F.D. on that table)

* How to find GIs using closure property:

ex

R (ABCD)

⇒ ABCD is C.K

A → B

B → C

C → D

D → A

2 →

keys :

length → 4 : ABCD

length → 3 : ABC ACD BCD ABD

length → 2 : AB AC AD BC BD CD

length → 1 : A B C D

→ if A^+ , B^+ , C^+ , D^+ are c.keys we do not need to check all other c.keys.

→ if A^+ , B^+ , C^+ , D^+ are c.keys then all other keys are c.keys if all are S.K. cannot be

→ we should start at level-1 (length=1).

A^+ : ABCDA ⇐ (A gives B

B^+ : BCDA B gives C

C^+ : DABC C gives D

D^+ : ABCD D gives A)

gate:

ex

R (ABCDEF)

C → F

E → A

EC → D

A → B

Find out which one is

candidate key?

options

(A) CD

(B) EC

(C) AE

(D) AC

$CD^+ : CDF$
 $EC^+ : ECDABF \leftarrow \text{All attr.}$
 $AE^+ : AEB$
 $AC^+ : ACFB$

Now, $E^+ : AB$ } NO S.K. therefore
 $C^+ : F$ } it is C.K.

$\Rightarrow$ 2nd Method (Shortcut):

$X^+ \Rightarrow (\overline{A}\overline{B}\overline{C}\overline{D}\overline{E}\overline{F})$
 $\uparrow$
 C.K.

$\Rightarrow$ look on right side of all FD
 and $FABD$ are derivable on left.
 CE are such attrⁿ which are not possible
 to identify.

$\therefore C \& E$ are must be part of left

Is CE^+ a key $(\underbrace{C^+, E^+}_{\text{are not S.K.}})$
 then only it is C.K.

eg

$R(EFGHIJ KLMN)$
 $EF \rightarrow G$
 $EH \rightarrow KL$
 $F \rightarrow IJ$
 $K \rightarrow M$
 $L \rightarrow N$

$\Rightarrow X^+ \Rightarrow E\overline{F}\overline{G}H\overline{I}\overline{J}\overline{K}\overline{L}\overline{M}\overline{N}$

EEH⁺ : EFHGKLIITMN

$$\left. \begin{array}{l} E^+ : E \\ F^+ : IJ \\ H^+ : H \end{array} \right\} \rightarrow \text{are not S.K. therefore it is C.K.}$$

Ans EFH

no need to check subset
if remove H $\Rightarrow$ EF we don't get H anywhere

if remove F $\Rightarrow$ EH not gives other attⁿ

if remove E $\Rightarrow$ FH not gives other attⁿ

eg

R(ABCD)

AB $\rightarrow$ CD

DA $\rightarrow$ A

Find out whether it gives C.K. or not.

$$X \Rightarrow R(ABCD)$$

$B^+ : B$ $\leftarrow$ not sufficient
C.K. must have B

$AB^+ : CDAB$ ✓

$BC^+ : BC$

$BD^+ : BDAC$ ✓

$ABC^+ : ABCD$ $\leftarrow$ can not be C.K.

it's subset is S.K.

$BCD^+ : BCDA$ $\leftarrow$ can not be C.K.

BD (subset) is S.K.

eg.

$R(AB C D E F)$

$AB \rightarrow C$

$C \rightarrow D$

$B \rightarrow AE$

$R(AB \overset{\vee}{C} \overset{\vee}{D} \overset{\vee}{E} F)$

$BF^+ : BF A E C D \leftarrow C.K.$

$B^+ : AE$

$F^+ : F$

eg.

$R(AB \overset{\vee}{C} \overset{\vee}{D})$

$AB \rightarrow CD$

$C \rightarrow A$

$D \rightarrow B$

(not applicable short cut key)

$AB^+ : ABCD \leftarrow C.K.$

$\rightarrow$ actual Method :

NORMALIZATION :

→ split table into multiple tables. is known as 'normalization.'

eg s_id, s_name, dob, c_id, c_name, prereq, duration, result_date, exam_result, marks, grade

→ Problem with this schema:

- Redundancy. (primary problem)
- Insert Anomaly (insertion problem)

↓

P.K SID, CID

if we add new course
SID will not be null.

(one of either can not be
null [SID, CID])

- Delete Anomaly (loss of data)
- Update Anomaly (occurs due to redundancy)

→ Solution:

levels : 1NF (1st Normal form)

↓

2NF (2nd Normal form)

↓

3NF (3rd Normal form)

↓

BCNF (Boyce-Codd Normal form)
(0% redundancy)

2 → $x \rightarrow y$ (BCNF)

When x is C.K. → x uniquely identify y .
S.K.

F.Ds ⇒ $sid \rightarrow sname, dob$

$cid \rightarrow cname, prereq, duration$

$sid, cid \rightarrow examdate, marks, grade$

$marks \rightarrow grade$

2 → 1NF → every attr. must have atomic domain.
(indivisible)

(This schema is by default 1NF -
according some assumption)

$sname, dob \rightarrow$ is atomic ←

- Relation is in first normal form if and only if all the attributes of the relation are atomic.

- We don't solve any problem in 1NF.

2 → A relation is in 2NF if and only if it satisfies 2 conditions

(i) It is already in 1NF.

(ii) No partial dependancy between key and non-key attributes. (prime & non-prime attributes)

$\Rightarrow$ c.k. $\Rightarrow$ sid, cid (prime att^s)
 - it is an att^s that is part of any c.k. is known as prime att^s.

- The att^s th which are not part of any c.k. is known as non prime att^s.

sid (prime) $\rightarrow$ sname, dob (non prime)

$p \rightarrow np$

$\uparrow$
 it depends upon

it is called

part of c.k.

partial dependency - ideally it depends upon whole c.k. (sid & cid)

- Therefore it is not in 2NF form

- 1st and 2nd f.d.s are partial dependency in given example.

- 3rd f.d. $\rightarrow$ full dependency

- 4th f.d. $\rightarrow$ np $\rightarrow$ np (no need to check 2nd condⁿ)

[ideally we want full dependency]

Solⁿ : split into sub-tables

(student) table - sid, sname, dob

(course) table - cid, cname, prereq, duration

(result) table - sid, cid, examdate, marks, grade

- for result relation,
 $\text{marks} \rightarrow \text{grade}$
 $\text{np} \rightarrow \text{np}$ } $\rightarrow$ redundancy

- split into two tables
 (1) sid, cid, examdate, marks (stu-marks)
 (2) marks, grade (stu-grades)

sid, cid $\rightarrow$ marks $\rightarrow$ grade } - ideally
 $A \rightarrow B$
 $B \rightarrow C \Rightarrow A \rightarrow C$ } there is no transitive depⁿ

2 $\rightarrow$ A relation is in 3NF if it follows two condⁿ
 (i) it is in 2NF
 (ii) No transitive dependency.

$\rightarrow$ for every $X \rightarrow A$, either X should be S.K or A should be prime attⁿ

[every left side should be S.K / CK
 right side should be prime attⁿ
 - It is in 3NF]

- split into 2 tables as above.

2 $\rightarrow$ BCNF : for every fd. left hand side should be S.K (NO redundancy)

In last 2 tables
 (i) sid, sname, dob
 (ii) cid, cname, prereq, duration

(iii) slid, cid, examdate, marks
(iv) marks, grade

eg Project-no, Project-nm, Empno, Empnm,
RateCategory, HrRate

→ FD
Project-no → project nm (Partial Dep)
Empno → Empnm, Ratecategory, HrRate (P.D)
Ratecategory → HrRate (np → np).

Project-no, Emp-no ← C.K

This is in 1NF.

⇒ In 2NF → $P \rightarrow np$

⇒ Split into subtables:

table-1 : project - details

project-no, project-nm

table-2 : emp - details

Emp-no, empnm, Rate category,
HrRate

table-3 : proj-emp

project-no, Emp-no

⇒ For 3NF : split emp details

emp-no, empnm, Rate category
and Rate-Hr

Ratecategory, HrRate

⇒ In BCNF:

proj no, proj name

emp no, emp name, RateCategory

RateCategory, HrRate

proj no, emp no.

eg $R(ABC) \Rightarrow$ split $R_1(A)$ & $R_2(BC)$ Randomly split tables

2 → Decomposition Method

	A	B	C
<u>R</u>	a ₁	b ₁	c ₁
	a ₂	b ₁	c ₁
	a ₁	b ₂	c ₂

$R_1(A)$ and $R_2(BC)$

↙

<u>R₁</u>	A
	a ₁
	a ₂

<u>R₂</u>	B	C
	b ₁	c ₁
	b ₂	c ₂

two types of decomposition (i) lossless
(ii) lossy (loss of data)

⇒ If we join two tables with natural join and if we get our original table then it is lossless decomposition.

→ No cross product

A	B	C
a ₁	b ₁	c ₁
a ₁	b ₂	c ₂
a ₂	b ₁	c ₁
a ₂	b ₂	c ₂

we don't get original table $\Leftarrow$ lossy decomposition
(more no. of tuples)

Now, $R_1(AB)$, $R_2(BC)$

R_1

A	B
a ₁	b ₁
a ₂	b ₁
a ₁	b ₂

R_2

B	C
b ₁	c ₁
b ₂	c ₂

cross product
(natural join)

A	B	C
a ₁	b ₁	c ₁
a ₂	b ₁	c ₁
a ₁	b ₂	c ₂

→ lossless decomposition :

If we replace R with R_1 and R_2 and we don't lose any data then it is called lossless decomposition.

- common attribute should be key in one of the tables.

Therefore, AB and B is key.

⇒ If we divide R into R_1 and R_2 then
 $(R_1 \cap R_2) \rightarrow R_1$

or
 $(R_1 \cap R_2) \rightarrow R_2$

$(AB \cap BC)$
 $B \rightarrow \underline{BC}$

* lossless join Property Algoⁿ:

$R (A_1, A_2, A_3, A_4, A_5)$

$R_1 (A_1, A_2, A_3, A_5)$

$R_2 (A_1, A_3, A_4)$

$R_3 (A_4, A_5)$

step:1 Create a matrix S where one row $i \Rightarrow$ ^{for} every relation i
 one column $j \Rightarrow$ each attribute j

	A_1	A_2	A_3	A_4	A_5
R_1	b_{11}	b_{12}	b_{13}	b_{14}	b_{15}
R_2	b_{21}	b_{22}	b_{23}	b_{24}	b_{25}
R_3	b_{31}	b_{32}	b_{33}	b_{34}	b_{35}

step:2 for every $S[i][j] = b_{ij}$

step:3 for each row i
 for each row j
 if R_i includes A_j
 $S(i,j) = a_j$

	A_1	A_2	A_3	A_4	A_5
R_1	a_1	a_2	a_3	b_{14}	a_5
R_2	a_1	b_{22}	a_3	a_4	b_{25}
R_3	b_{31}	b_{32}	b_{33}	a_4	a_5

step 4

$A_1 \rightarrow A_3 A_5$
 $A_5 \rightarrow A_1 A_4$
 $A_3, A_4 \rightarrow A_2$

→ f.d.s

compulsory to check

take each fd

(X) (Y)
 $A_1 \rightarrow A_3 A_5$

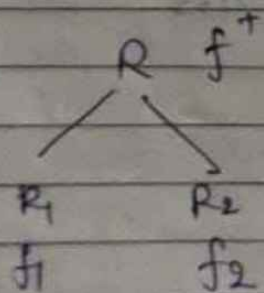
↑
 which rows are having same value

	<u>A₁</u>	A ₂	<u>A₃</u>	<u>A₄</u>	<u>A₅</u>
R ₁	a ₁	a ₂	a ₃	a ₄	a ₅
R ₂	a ₁	b ₂₂	a ₃	a ₄	b ₂₅ → a ₅
R ₃	b ₃₁	b ₃₂	b ₃₃	a ₄	a ₅

now $A_5 \rightarrow A_1 A_4$

⇒ If there any cell 'a's in one row then it is lossless decomposition.

⇒ F.D. preservation :



$$(f_1 \cup f_2)^+ \subseteq f^+$$

Q1
 $R(ABCDE)$
 $P: \{AB \rightarrow C, C \rightarrow E, B \rightarrow D, E \rightarrow A\}$
 $R_1(BCD)$
 $R_2(ACE)$

⇒ Dependency Preservation:

For each FD, $\alpha \rightarrow \beta$

result - α

repeat

for each R_i in decomposition
 $t = (\text{result} \cap R_i)^+ \cap R_i$
 $\text{result} = \text{result} \cup t$
 until (result does not change)
 result should contain all attr. of β .

[$AB \rightarrow C$ result = AB
 $X \rightarrow Y$ $R_1(BCD)$
 $t = (AB \cap BCD)^+ \cap BCD$
 $= B^+ \cap BCD$
 $= BD \cap BCD = BD$
 $\text{result} = AB \cup BD = ABD$
 $R_2(ACE)$
 $t = (ABD \cap ACE)^+ \cap ACE$
 $= A^+ \cap ACE$
 $= A \cap ACE = A$
 $\text{result} = ABD \cup A = ABD$

Now ABD does not contain C
 we lose $AB \rightarrow C$ dependency
 therefore it doesn't preserve depⁿ.]

eg. $R(ABC)$ Decompositions $R_1(AB)$
 $A \rightarrow B$ $R_2(BC)$
 $B \rightarrow C$
 $C \rightarrow A$

→ lossless + Depⁿ preservation

* Canonical Cover:

from $f \Rightarrow$ we find f_1
 $f_1 \Rightarrow$ less no. of FD.

eg $f \{$

$AB \rightarrow CD$

$A \rightarrow E$

$E \rightarrow C \}$

- it gives same closure as the original contain.

$R \Rightarrow$ Relation schema

$f \Rightarrow$ set of FD on R

for

Right side

$A \in \beta$ and A is extraneous

$f' = (f - \{\alpha \rightarrow \beta\}) \cup (\alpha \rightarrow (\beta - A))$

α^+ on f'

$\alpha^+ \Rightarrow A$

$f' \{ AB \rightarrow D \}$ remaining 2 are same

for

left side

$A \in \alpha$, and A is extraneous

let $\gamma = \alpha - \{A\}$

γ^+ on f

$\gamma^+ \Rightarrow \beta$ contains all attr^s of β

A is extra

$AB \rightarrow C$

$AB \rightarrow D$

$A \rightarrow E$

$A \rightarrow E$

$E \rightarrow C$

$E \rightarrow C$

canonical cover

eg

$A \rightarrow BC$

$B \rightarrow C$

$A \rightarrow B$

$AB \rightarrow C$